

doc link:-

https://docs.google.com/document/d/1ByEHwNMeRgl9OI_fTwRn6jY4leHdX6Q2kFKmTgZ4oQg/edit?usp=sharing

chapter link:- <https://jax-ml.github.io/scaling-book/transformers/>

Question 1: How many parameters does a model with $D=4096$ $F=4 \cdot D$ $V=32,000$ and $L=64$ have? What fraction of these are attention parameters? How large are our KV caches per token? You can assume $N \cdot H=D$ and multi-head attention with int8 KVs.

My answer:-

part 1)

kvq parameters= $3DNH$ (3 W_k, W_q, W_v matrix)

attention out parameter= NHD (convert kvq processed to D form removes nH)

MLP num parameter= $3DF$ (W_{in1} for weightg feationg through gelu selection, w_{in2} for featurg, w_{out} to reconvert to desirable form)

vocab= VD (vocab)

num layers of mlp repeat = L

number of parameters= $VD+L*(4*NDH+DF)=16B$ parameters

part 2) fraction= $(kvq \text{ parameters} + \text{attention out parameter}) / (\text{total}) = 1/4$

part 3)

full kv= $2(\text{for } k \text{ and } v)L(\text{num layers})T(\text{tokens})NH(\text{num parameter for 1 token})=512kb$ per token

Question 2: How many total FLOPs are required to perform $A[B_X, D_Y] * W[D_Y, F]$ on $\{ 'X': 4, 'Y': 8, 'Z': 4 \}$. How many FLOPs are performed by each TPU?

My answer:-

total flops is $2BDFZ$ and per tpu is $2bdf/xy$

Question 3: How many FLOPs are involved in performing
 $A[I,J,K,L] * B[I,J,M,N,O] \rightarrow C[K,L,M,N,O]$?

My answer:-
 $2 * IJKLMNO$

Question 4: What is the arithmetic intensity of self-attention (ignoring the Q/K/V/O projections)? Give the answer as a function of the Q and KV lengths T and S. At what context length is attention FLOPs-bound? Given the HBM bandwidth of our TPUs, plot the effective relative cost of attention to the FFW block as the context length grows.

My answer:-
 bytes key=q=v=BTNH so total space=3BTNH
 now k and v operation of (BTNH)X(BTNH)→(BTTN) is of $BNHT^2$
 and attention to value is (BTTN)X(BTNH)→(BTNH) is BT^2NH
 and total flops is $2BNHT^2$
 so AI is proportional to T and around at T as sequence length is equal to input length
 around
 so around at $T=240$

Question 5: At what sequence length are self-attention FLOPs equal to the QKVO projection FLOPs?

My answer:-
 attention flops is $2BNHTT$
 projection flops for kqv is (BTD)X(DNH)=BTDNH so total $3BTDNH$
 and for O is $BTNH * NHD$ so it is $BTNHD$ so total $4BTNHD$
 so at $T=2D$

Question 6: Say we only save the output of each of the 7 main matmuls in a Transformer layer during our forward pass (Q, K, V, O + the three FFW matrices). How many extra FLOPs do we need to “rematerialize” during the backwards pass?

my answer:-
 so we have saved $4BTNHD$ and $3BTF$

extra is attention creation and applying and element wise MLP and MLP projection

so it is $(2 \times \text{BNHTT} + \text{BTF} + \text{BTfD}) \times 2$ flops operations

Question 7: DeepSeek v3 says it was trained for 2.79M H800 hours on 14.8T tokens ([source](#)). Given that it has 37B activated parameters, roughly what hardware utilization did they achieve? *Hint: note that they used FP8 FLOPs without structured sparsity.*

My answer:-

rough flops required is 6 num parameters * tokens

1.513×10^{15} FLOPs/s is the s without spars performance
so the time compute done is $1.513 \times 10^{15} \times 2.79 \times 10^6 \times 60 \times 60$

and the compute required is $6 \times 2.76 \times 10^6 \times 1.48 \times 10^{12}$

so the efficienct is 20 percentage

Question 8: Mixture of Experts (MoE) models have E copies of a standard dense MLP block, and each token activates k of these experts. What batch size in tokens is required to be compute-bound for an MoE with weights in int8 on TPU v5e? For DeepSeek, which has 256 (routed) experts and $k=8$, what is this number?

My answer:-

each expert is mlp layer and mlp have flops of 6BTfD flops and it loads 3DF matrices

so per token activation computation is 6BFD and loading same model 3DF and comput k times and load all E so

$B > E/2k \times 240$

so for deep seek its 3840

